

PES University - Dept. of CSE

Lab 1: Use-Case Flow Specification

PROBLEM STATEMENT #43: Feature Flag & Dynamic Config Manager

Use Case ID:	UC-01
Use Case Name:	Evaluate Feature Flag & Dynamic Config
Primary Actor:	Client SDK / Application (also Software Engineer)
Preconditions:	1. Client SDK is initialized with valid API key. 2. Flag targeting rules are active in system database.
Postconditions:	1. System evaluates & returns boolean/JSON flag payload. 2. Telemetry event is recorded.

1. Main Success Scenario

1. Client SDK sends an evaluation request payload specifying flag_key, user_id, and user_attributes over HTTP/gRPC.
2. System validates client request header, verifying API key authenticity and environment permissions.
3. System retrieves active feature flag schema and rollout rules for the target environment.
4. System evaluates environment-specific overrides (e.g., Development vs. Production).
5. System evaluates user attribute targeting rules (e.g., country code, beta tester status).
6. System hashes user_id using consistent hashing (MurmurHash3) to derive rollout bucket (0–100%).
7. System compares user bucket percentage against configured rollout threshold to determine flag state.
8. System caches evaluated flag state in edge memory (Redis) for rapid sub-millisecond future lookups.
9. System returns HTTP 200 OK containing boolean state, variation key, and evaluation metadata.
10. Client SDK receives response, updates internal state, and host app executes target feature branch.

2. Alternate Flow (2a: Network Timeout / Fallback to Local Cache)

- 2a1. Client SDK detects network socket timeout (>50ms) or central API service returns HTTP 5xx error.
- 2a2. Client SDK catches the timeout exception without interrupting host application execution thread.
- 2a3. Client SDK retrieves the most recent valid flag state from its persistent local disk/memory cache.
- 2a4. If local cache is empty, SDK defaults to system hardcoded fallback flag value defined by developer.
- 2a5. SDK logs offline warning telemetry event and schedules background sync retry.
- 2a6. Use case completes successfully with fallback state.